

C.4 General Approaches to Parameter Estimation

Until this point, we have used the sample average to illustrate the finite and large sample properties of estimators. It is natural to ask: Are there general approaches to estimation that produce estimators with good properties, such as unbiasedness, consistency, and efficiency?

The answer is yes. A detailed treatment of various approaches to estimation is beyond the scope of this text; here, we provide only an informal discussion. A thorough discussion is given in Larsen and Marx (1986, Chapter 5).

Method of Moments

Given a parameter θ appearing in a population distribution, there are usually many ways to obtain unbiased and consistent estimators of θ . Trying all different possibilities and comparing them on the basis of the criteria in Sections C.2 and C.3 is not practical. Fortunately, some methods have been shown to have good general properties, and, for the most part, the logic behind them is intuitively appealing.

In the previous sections, we have studied the sample average as an unbiased estimator of the population average and the sample variance as an unbiased estimator of the population variance. These estimators are examples of **method of moments** estimators. Generally, method of moments estimation proceeds as follows. The parameter θ is shown to be related to some expected value in the distribution of Y , usually $E(Y)$ or $E(Y^2)$ (although more exotic choices are sometimes used). Suppose, for example, that the parameter of interest, θ , is related to the population mean as $\theta = g(\mu)$ for some function g . Because the sample average $\bar{Y}$ is an unbiased and consistent estimator of μ , it is natural to replace μ with $\bar{Y}$, which gives us the estimator $g(\bar{Y})$ of θ . The estimator $g(\bar{Y})$ is consistent for θ , and if $g(\mu)$ is a linear function of μ , then $g(\bar{Y})$ is unbiased as well. What we have done is replace the population moment, μ , with its sample counterpart, $\bar{Y}$. This is where the name “method of moments” comes from.

We cover two additional method of moments estimators that will be useful for our discussion of regression analysis. Recall that the covariance between two random variables X and Y is defined as $\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)]$. The method of moments suggests estimating σ_{XY} by $n^{-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$. This is a consistent estimator of σ_{XY} , but it turns out to be biased for essentially the same reason that the sample variance is biased if n , rather than $n - 1$, is used as the divisor. The **sample covariance** is defined as

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}). \quad \text{C.14}$$

It can be shown that this is an unbiased estimator of σ_{XY} . (Replacing n with $n - 1$ makes no difference as the sample size grows indefinitely, so this estimator is still consistent.)

As we discussed in Section B.4, the covariance between two variables is often difficult to interpret. Usually, we are more interested in correlation. Because the population correlation is $\rho_{XY} = \sigma_{XY}/(\sigma_X \sigma_Y)$, the method of moments suggests estimating ρ_{XY} as

$$R_{XY} = \frac{S_{XY}}{S_X S_Y} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\left(\sum_{i=1}^n (X_i - \bar{X})^2 \right)^{1/2} \left(\sum_{i=1}^n (Y_i - \bar{Y})^2 \right)^{1/2}}, \quad \text{C.15}$$

which is called the **sample correlation coefficient** (or *sample correlation* for short). Notice that we have canceled the division by $n - 1$ in the sample covariance and the sample standard deviations. In fact, we could divide each of these by n , and we would arrive at the same final formula.

It can be shown that the sample correlation coefficient is always in the interval $[-1, 1]$, as it should be. Because S_{XY} , S_X , and S_Y are consistent for the corresponding population parameter, R_{XY} is a consistent estimator of the population correlation, ρ_{XY} . However, R_{XY} is a biased estimator for two reasons. First, S_X and S_Y are biased estimators of σ_X and σ_Y , respectively. Second, R_{XY} is a ratio of estimators, so it would not be unbiased, even if S_X and S_Y were. For our purposes, this is not important, although the fact that no unbiased estimator of ρ_{XY} exists is a classical result in mathematical statistics.

Maximum Likelihood

Another general approach to estimation is the method of *maximum likelihood*, a topic covered in many introductory statistics courses. A brief summary in the simplest case will suffice here. Let $\{Y_1, Y_2, \dots, Y_n\}$ be a random sample from the population distribution $f(y; \theta)$. Because of the random sampling assumption, the joint distribution of $\{Y_1, Y_2, \dots, Y_n\}$ is simply the product of the densities: $f(y_1; \theta)f(y_2; \theta) \cdots f(y_n; \theta)$. In the discrete case, this is $P(Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n)$. Now, define the *likelihood function* as

$$L(\theta; Y_1, \dots, Y_n) = f(Y_1; \theta)f(Y_2; \theta) \cdots f(Y_n; \theta),$$

which is a random variable because it depends on the outcome of the random sample $\{Y_1, Y_2, \dots, Y_n\}$. The **maximum likelihood estimator** of θ , call it W , is the value of θ that maximizes the likelihood function. (This is why we write L as a function of θ , followed by the random sample.) Clearly, this value depends on the random sample. The maximum likelihood principle says that, out of all the possible values for θ , the value that makes the likelihood of the observed data largest should be chosen. Intuitively, this is a reasonable approach to estimating θ .

Usually, it is more convenient to work with the *log-likelihood function*, which is obtained by taking the natural log of the likelihood function:

$$\log [L(\theta; Y_1, \dots, Y_n)] = \sum_{i=1}^n \log [f(Y_i; \theta)],$$

C.16

where we use the fact that the log of the product is the sum of the logs. Because (C.16) is the sum of independent, identically distributed random variables, analyzing estimators that come from (C.16) is relatively easy.

Maximum likelihood estimation (MLE) is usually consistent and sometimes unbiased. But so are many other estimators. The widespread appeal of MLE is that it is generally the most asymptotically efficient estimator when the population model $f(y; \theta)$ is correctly specified. In addition, the MLE is sometimes the **minimum variance unbiased estimator**; that is, it has the smallest variance among all unbiased estimators of θ . [See Larsen and Marx (1986, Chapter 5) for verification of these claims.]

In Chapter 17, we will need maximum likelihood to estimate the parameters of more advanced econometric models. In econometrics, we are almost always interested in the distribution of Y conditional on a set of explanatory variables, say, $X_1, X_2, \dots, X_k$. Then, we replace the density in (C.16) with $f(Y_i | X_{i1}, \dots, X_{ik}; \theta_1, \dots, \theta_p)$, where this density is

allowed to depend on p parameters, $\theta_1, \dots, \theta_p$. Fortunately, for successful application of maximum likelihood methods, we do not need to delve much into the computational issues or the large-sample statistical theory. Wooldridge (2002, Chapter 13) covers the theory of maximum likelihood estimation.

Least Squares

A third kind of estimator, and one that plays a major role throughout the text, is called a **least squares estimator**. We have already seen an example of least squares: the sample mean, $\bar{Y}$, is a least squares estimator of the population mean, μ . We already know $\bar{Y}$ is a method of moments estimator. What makes it a least squares estimator? It can be shown that the value of m that makes the sum of squared deviations

$$\sum_{i=1}^n (Y_i - m)^2$$

as small as possible is $m = \bar{Y}$. Showing this is not difficult, but we omit the algebra.

For some important distributions, including the normal and the Bernoulli, the sample average $\bar{Y}$ is also the maximum likelihood estimator of the population mean μ . Thus, the principles of least squares, method of moments, and maximum likelihood often result in the *same* estimator. In other cases, the estimators are similar but not identical.

C.5 Interval Estimation and Confidence Intervals

The Nature of Interval Estimation

A point estimate obtained from a particular sample does not, by itself, provide enough information for testing economic theories or for informing policy discussions. A point estimate may be the researcher's best guess at the population value, but, by its nature, it provides no information about how close the estimate is "likely" to be to the population parameter. As an example, suppose a researcher reports, on the basis of a random sample of workers, that job training grants increase hourly wage by 6.4%. How are we to know whether or not this is close to the effect in the population of workers who could have been trained? Because we do not know the population value, we cannot know how close an estimate is for a particular sample. However, we can make statements involving probabilities, and this is where interval estimation comes in.

We already know one way of assessing the uncertainty in an estimator: find its sampling standard deviation. Reporting the standard deviation of the estimator, along with the point estimate, provides some information on the accuracy of our estimate. However, even if the problem of the standard deviation's dependence on unknown population parameters is ignored, reporting the standard deviation along with the point estimate makes no direct statement about where the population value is likely to lie in relation to the estimate. This limitation is overcome by constructing a **confidence interval**.

We illustrate the concept of a confidence interval with an example. Suppose the population has a $\text{Normal}(\mu, 1)$ distribution and let $\{Y_1, \dots, Y_n\}$ be a random sample from this population. (We assume that the variance of the population is known and equal to unity for the sake of illustration; we then show what to do in the more realistic case that the variance